Od jezyčnogo kanona do povtorimoj publikacije: metodologija mašinno-pomaganoj medžuslovjanskoj redakcije

From a language canon to a reproducible publication: machine-assisted Interslavic editing of Emmy Noether's mathematical works

Machine-assisted Interslavic editorial project

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Abstract

Medžuslovjansky. Tuta statja dokumentuje tvorjenje polnogo medžuslovjanskogo izdanja matematičnyh děl Emmy Noether v redakcijnoj latinici i deterministično izvedenoj kirilici. Cělj jest praktično čitanje govoriteljami slovjanskyh jezykov, uključajuči čitateljev bez anglijskogo jezyka, a ne sama demonstracija jezyčnogo eksperimenta. Izdanje v020 ne prěpisuje latiničny kanon: ono popravjaje role, ktore upravljajut izvedenjem kirilice. Bylo pregledano vseh 4 560 naslědženyh zapisov rolij po pojavjenju; popravljeno bylo 364 klasifikacij i dodano 8 nedostajučih granic polnyh neizměnjenyh identitetov. Projektor v7 dokazano izvršaje vseh 372 efektov: 346 ohranjajut latinny tekst, a 26 projicirajut go v kirilicu. Skutočna izhodna razlika sodrži 125 linij vnutri licencovanogo množstva od 309 linij i nulu izměn izvan njega. Dva polna čitateljnika imajut 565 i 588 stranic; relizna kontrola imaje 38 uspěšnyh prověrok i nulu pogrěšek. "Nula nejasnostij" znači samo, že v manifestu ne ostalo neršenyh klasifikacij rolij. Ona ne znači rodno-govoriteljsku, obćinsku ili naučnu certifikaciju. Vse iztočniki, rěšenja, programy, kontrolne sumy, ograničenja i javne adresy sut ostavljene tako, da druga sesija ili recenzent može povtoriti, kritikovati ili monotono prodolžiti izdanje.

English. This paper documents the production of a complete Interslavic edition of Emmy Noether's mathematical works in editorial Latin script and deterministically derived Cyrillic. The purpose is practical reading by speakers of Slavic languages, including readers without English, rather than a display of linguistic experimentation. Release v020 does not rewrite the Latin canon; it corrects the roles that govern Cyrillic derivation. All 4,560 inherited role records were reviewed occurrence by occurrence; 364 classifications were corrected and 8 missing boundaries for complete uninflected identities were added. Projector v7 demonstrably realizes all 372 effects: 346 preserve Latin material and 26 project it into Cyrillic. The actual output delta comprises 125 lines within a licensed set of 309 lines and zero changes outside it. The two complete readers contain 565 and 588 pages; release QA reports 38 passing checks and zero errors. "Zero ambiguity" means only that the role manifest contains no unresolved classifications. It is not native-speaker, community, or scholarly certification. Sources, decisions, software, hashes, limitations, and public access points are retained so that another session or reviewer can reproduce, challenge, or monotonically extend the edition.

Čest I - Polny medžuslovjansky tekst

1 Predmet, cělj i čitatelji

Predmet jest jedno polno medžuslovjansko jezyčno izdanje ograničenogo korpusa Emmy Noether: numerovane Statje 1–43, lekcijno dělo 44, Statja 45 i bibliografija. To ne jest antologija priměrov i ne jest zbirka nevezanyh transkripcij. Javny čitateljnik dava dva pisemne vida jedinogo jezyčnogo objekta:

- `isv-Latn` jest kanoničny redakcijny jezyčny tekst;
- `isv-Cyrl` jest deterministična projekcija togo teksta, a ne nezavisny svědok prěvoda.

Toto razděljenje jest metodologično važno. Izvedena azbuka ne može sama potvrditi pravilnost prěvoda, iz ktorego jest stvorjena. Jezyčne popravky vhodet jedino v latinny kanon; kirilica mora byti ponovno izvedena, testovana, sgrađena i vizualno prověrjena po zamrznutju vhodnyh bajtov. V020 jest osobito jasny slučaj: latinny tekst ostavaje bajt-po-bajtu ravny v019, ale izvedena kirilica se popravjaje, ibo pregled rolij odkryl pogrěšne granice medžu jezyčnym tekstom, imenami, citatami i rimskymi označenjami.

Praktična heuristika jezyčnogo napredka ne jest pytanje “koliko tekst izgleda slovjansky?” Uporabjajemo sojedinjenje četyrěh pytanj:

1. Može li govoritelj različnogo slovjanskogo jezyka pravděpodobno razobrati větu bez anglijskogo teksta?
2. Jest li izbrana forma podprta deklarovanym medžuslovjanskim slovnikom ili gramatikoju, albo jasno zapisanym domovym pravilom projekta?
3. Ohranjaje li forma točny matematičny predmet, kvantifikaciju i odnos argumentov německogo iztočnika?
4. Jest li vsaky slučaj izključka, identiteta, citaty ili nejasnosti reprezentovany povtorimoju roloju, vměsto bezglasnogo izostavjenja?

Zato izdanje jest najbolje razuměti kako popravljivo naučno rabotno izdanje: javno, pregledljivo, vezano kontrolnymi sumami i otvorjeno za uločšenje, ale bez tvrdženja o rodno-govoriteljskoj ili kritično-izdateljskoj certifikaciji.

2 Arhitektura avtoriteta i kustodije

Metoda razděljaje iztočny avtoritet, jezyčny kanon, izvedenu azbuku, redakcijne dokazy i javne produkte. Nijedin sloj ne smije samovoljno prěvzeti funkciju drugogo.

Table 1: Role avtoriteta v NOETHER-ISV-v020.

Rola	Upravljajuči objekt	Poslědstvo
Německy iztočny avtoritet	Naslědženy objekt `NOETH-DE-ED-0015`, pokazatelj `NOETH-DE-AUTH-v052-20260815`	Upravjaje iztočnym smyslom, strukturoju i matematičnym sravnjenjem; medžuslovjanska linija go ne rediguje.
Latiničny jezyčny kanon	Četyri `source_latin/*.tex` fajly pri glave `ISV019-EDIT-0171`	Jedino město jezyčnyh redakcij. V020 ne měnjaje nijedin jego bajt.
Manifest rolij	`ISV020-REVIEWED-PROJECTION-ROLE-MANIFEST.json`	Veže 4 560 pregledanyh zapisov, 8 dodatkovyh granic i nulu neréšenyh klasifikacij k točnym iztočnym lokatoram.
Kirilična površina	Projektor v7 nad točnym latinnym vektorom i manifestom	Povtorimo tvorjeny izhod; ne može nadglasovati latinny kanon niti nezavisno potvrditi prěvod.
Relizne produkty	Dva čitateljnika, vezany čitateljnik, dva ZIP-arhiva i SHA-256 manifest	Člověku dostupno razprostranjenje; produkt ne može redefinovati kanon, iz ktorego byl stvorjen.

Německy objekt naslědženy iz v019 imaje SHA-256 `51A25101C04877AE740989E72B2AD65A7A7E65B081077C4A518BF1737AD5B907`. Ta identiteta označaje kontrolny iztočnik prěma kojemu latinny kanon byl zamrznuty; ona ne jest tvrdženje, že projekt publikoval novy kritičny německy tekst. Globalna mnogojezyčna zbirka jest povezana odnosom `IsPartOf` s 10.5281/zenodo.20412587, a identificirany německy objekt odnosom `IsTranslationOf`.

3 Mašinno čitljivo odkryvanje kanona

Povtorimost začinaje s odkryvanjem. Buduća sesija ne smije ugadyvati ktory iz podobnyh fajlov jest kanon. Zato koren repozitorija imaje `CANON_INDEX.json`, ktory deklaruje:

- jezyk i BCP 47 označenja pisma;
- četyri kanonične latinne fajly i četyri izvedene kirilične fajly;
- točne bajtove i SHA-256 vsakej upravljajučej jedinice;
- latinnu redakcijnu glavu i identitet žurnala, do ktorogo se može samo dodavati;
- manifest rolij, projektor, test i prověrčny dokument;
- německy avtoritet, granice pisanja i granice dozvoljenyh tvrdženj;
- deterministične komandy sgrađanja i imena javnyh artefaktov.

Indeks jest deklarativny, a ne samopotvrđajuči. On govori agentu čto objekty *sut*; izvršive prověrky dokazujut, jest li deklarovana identiteta ješče pravdiva. Tuto razděljenje odklanjaje kružno pravilo "najnověje izgledajuči fajl jest kanon" i dava osnovu, k ktoroj druga transkripcija, prěvod ili azbučna projekcija može bezpečno pridati novy sloj.

4 Čto AI děla i čto ne može rěšiti samo

Projekt uporablja AI kako redakcijny instrument, ne kako iztočny avtoritet. Mašinny trud obimaje sravnjenje větnyh kontekstov, tvorjenje kandidatnyh inventarov, klasifikaciju pojavjenj, formulovanje alternativ, generovanje strukturnyh testov i tvorjenje javnoj dokumentacije. Deterministične programy potom autentifikuju vhody, priměnjajut samo zamrznute rěšenja i odklanjajut nepredvidženu izměnu.

Metoda ima slědujuči cikl:

1. zamrznuti iztočny i ciljevy bajtovy vektor;
2. najdti materialne jezyčne ili rolne kandidaty bez avtomatičnogo globalnogo zaměnjenja;
3. pregledati vsako pojavjenje v větnom, bibliografičnom, matematičnom i TeX-kontekstu;
4. zapisati izbranu rolu, dokaz, odklonjene alternativy, neuvěrjenost i poslědstva;
5. napraviti manifest ili redakcijno rěšenje, vezano k točnomu lokatoru i kontrolnoj sumě;
6. izvesti izhod čistym programom i dokazati dopuščeny obseg razlike;
7. sgraditi, iztegnuti tekst, prověriti fonty, navigaciju, strukturu i vizualne stranice;
8. publikovati izvorniki, dokazy i haše zajedno s čitateljnikom, potom anonimno pročitati javne bajty nazad.

AI-prědlog ne stavaje kanonom zato, že zvuči uvěrlivo. On stavaje dopustimym samo kogda jest lokalizovany, podporjen deklarovanym dokazom, ograničeny v obsegu i povtorimo prověrjen. Kde dokaz ne jest dostatočny, pravilen rezultat jest ograničeno tvrdženje ili zapisano neréšeno pytanje, a ne izmišljena certifikacija.

5 Polny pregled rolij i korekcija v020

V019 već imalo funkcionalnu kiriličnu projekciju, ale naslědženy inventar byl stvorjen iz kandidatskyh skenerov, ktore mogut smatrati inojezyčno izgledajuči token, stilizovany tekst ili jedin znak za avtomatično ohranjajemy material. V020 zato je pregledalo vse 4 560 zapisy po pojavjenju, ne samo evidentne pogrěšky.

Kromě 364 popravjenyh klasifikacij, pregled je dodal 8 nedostajučih rasponov za polne neizměnjene identitety. Tako obstajajut 372 vezane efekty. Projektor v7 prověrjaje vsaky efekt odvojeno: 346 rasponov mora ohraniti latinny tekst i 26 mora go projicirati. Nula efektov ostavaje neprověrjena.

Licenca izměn ne jest slobodna lista cijelyh fajlov. Manifest izračunava 309 linij, na ktoryh nějakovy pregledany efekt može zakonno izměniti izhod. Skutočna razlika imaje 125 linij: 0 v dělu 44, 6 v statji 45, 115 v statjah 1–43 i 4 v bibliografiji. Nijedna linija se ne měnjaje izvan licencovanogo množstva. Četyri latinne izvorniky imajut nulu izměnjenyh bajtov.

Status "nula nejasnostij" treba čitati operacionno: vsaky zapis inventara imaje jednu završenu projekcijnu

Table 2: Pregled klasifikacij rolij.

Kategorija	Pregledane zapisy	Popravjene klasifikacije
Stilizovane površiny	883	1
Kandidaty imen	616	0
Pojavjenja inojezyčnogo leksikona	2 371	270
Vidime jedno-bukvene tokeny	690	93
Vse skupaj	4 560	364

rolu i točny lokator. On ne govori, že prirodny jezyk ne imaje polisemije, že vsaky govoritelj izbere tu samu formu, ili že někoj rodny govoritelj pregledal cijely korpus.

6 Jezyčna normalizacija latinnogo kanona s žurnalom, do ktorogo se može samo dodavati

Latinna glava naslědžuje 171 materialnyh rěšenj iz v019. Rěšenje jest dopustimo samo ako imenuje prědhodnika, točnu ciljnu poziciju, prědlagany naslědnik, relevantny avtoritet, odklonjene alternativy, motivaciju, neuvěrjenost i poslědnu prověrku. Aplikator autentifikuje vsaky vhod i odklanjaje mnogoznačny cilj. V žurnal se može samo dodavati: pogrěšny ili pozdněje zaměnjeny postup ne izčezaje, ale ostavaje čitljivym s novym poslědnikom.

Glavne jezyčne klasy sut:

- **Lažni prijatelji i smyslove razděljenja.** Podobnost s ruskym, ukrajinskym, poljskym, češskym ili južnoslovjanskym oblikom sama ne dostaje. Obstajanje, sučnost, sostav, sodržanje, prědpoloženje, poslědstvo i odgovaranje sut rěšene po větnom i matematičnom smyslu.
- **Morfologija i větny položaj.** Soglasovanje, kosy padeži, genitiv množiny, pričestja, glagolne koncovky i poredkove formy sut klasifikovane prěd popravkoju. Globalna zaměna koncovky jest odklonjena, ibo formuly, imena i citaty mogut imitovati običny morfem.
- **Vezy i logika dokaza.** Uzročne, zaključne, uslovne, ekvivalentne, protivne i apodozne vezy sut sravnjene s odnosom iztočnyh klauz. Čestost slova ne smije obrnuti implikaciju ili oslabiti kvantifikator.
- **Imena, naslovi i citaty.** Polny neizměnjeny identitet i izvornojezyčna bibliografična površina sut ohranjene. Integrirana ili sklanjana forma projiciraje se kogda funkcionuje kako medžuslovjanska morfologija.

V020 ne otvara te latinne rěšenja ponovno. Ono popravjaje samo mašinno tolkovanje, ktory časti zamrznutogo teksta sut vidimy medžuslovjansky tekst, a ktore sut zaštićene identitete ili struktura.

7 Matematična i TeX-věrnost

Prirodnojezyčna normalizacija jest podredžena matematičnoj identitetě. Zaštićene granice obimajut matematičny režim, imena komand, opcije komand, oznaky, odkazy, ključe citat, puti, komentare i dokazano izvornojezyčne argumenty. Za vsaky latinni/kiriličny par odstranjenje Unicode-bukv i kombinovanyh znakov dava tu samu ne-bukvenu struju. Taka prověrka ohranjaje zagrady, kontrolne slova, operatory, čisla, poravnateljne znaky i matematične symbole.

Ranějši neuspěh pokazuje, začto jezyčny skener ne dostaje. Projektor v5 je pogrěšno čital TeX-jedinicu `24em` kako vidimo slovo i stvoril kiriličnu jedinicu, ktora ne jest zakonny TeX. Projektor v6 je zato dobil eksplicitnu gramatiku numeričnyh razměrov, a v7 ju naslědžuje i ponovno prověrjaje. Ohranjeno jest 447 raněje izloženih pojavjenj: 219 `em` i 228 `pt`; kiriličny ostatok razměrne jedinice jest nula. Neuspěh ostavaje v dokazah, ibo “čisty” leksičny rezultat ne dokazuje izvršivu pravilnost iztočnogo programa.

8 Deterministični izvod v7

V7 jest čista funkcija četyrěh zamrznutyh latinnyh fajlov, pregledanogo manifesta rolij i pinovanyh zavisnostij. On:

1. autentifikuje razměr i SHA-256 vsakoj latiničnoj jedinice;
2. prověrjaje točnu topologiju iztočnikov i manifesta;
3. razděljuje TeX na vidime i zaštićene role;
4. sjedinjuje pregledane granice imen, citat, rimskyh označenj i razměrov;
5. projiciraje ostale vidime riječi zamrznutym medžuslovjanskim transliteratorom;
6. nezavisno rekonstruuje izhod s tragom i zahtěvaje bajtnu identitetu dvoh izvodov;
7. dokazuje vsaky iz 372 efektov i vsaku izměnjenu liniju;
8. piše izhody transakcijno samo poslě vseh prověrok.

Hostilny testovi naměrno měnjajut iztočny bajt, izostavljajut jedinicu topologije, prědlagajut nebezpečny cilj izhoda, kvarjajut pin manifesta i kvarjajut pin starogo izhoda. Vse pet slučajev bylo odklonjeno prě pisanjem. Dva nezavisna izračuna dajut bajt-po-bajtu ravne četyri kirilične izhody. Test i projektor sami sut hašovani, tako že “isti postup” znači identičny program, ne samo podobno opisanje.

9 Sgrađanje i mnogoslojna kontrola

Vse XeLaTeX-operacije tečut serijno, s dvoma prohodami po vsakom dokumentu, s izključenym shell escape i fiksovanym `SOURCE_DATE_EPOCH`. Osm redaktabilnyh komponentov davajut dva polna čitateljnika.

Table 3: Prověrjene komponentne čitateljniki v020.

Artefakt	Stranice	Bajty	SHA-256
Latiničny čitateljnik	565	3 000 522	`B27278EA138724ED258CFE5089CABA70A7 9C640D0B71CCD9836CA3C8CB8BCD13`
Kiriličny čitateljnik	588	3 300 221	`4AE9847672AB53F1D5B54468ED9B202D45 EF5969C7A34B1CE380B28C6516DB29`

Relizna kontrola imaje 38 uspěšnyh prověrok i nulu pogrěšek. Ona obimaje točno povratno čitanje iztočnikov, TeX i ne-bukvenu strukturu, fatalne žurnale, straničnu topologiju, fonty, `pdftotext`, popis pisem i identitet teksta medžu komponentami i kumulativnymi čitateljnikami. Vezany javny čitateljnik mora imati 6 uvodnyh/razdělnyh stranic plus 565 latiničnyh i 588 kiriličnyh stranic, vkupno 1 159. Njegov graditelj odklanjaje drugu topologiju.

Strukturny test ne dokazuje krasivy izgled. Zato finalna relizna kontrola takože zahtěvaje serijno renderovanje i vizualny pregled novogo uvoda, vsakej izměnjene stranice i susědnyh kontrolnyh stranic, plus razdělnyh granic dvoh pisem. Tekstny izteg ne može sam odkryti prěkryvanje, obrezanje ili nečitljivu glifnu zaměnu.

10 Publikacija, dostup i ohranjanje

Javna površina Zenodo imaje četyri produkta:

1. `00_NOETHER_INTERSLAVIC_COMPLETE_LINKED_READER.pdf`;
2. `01_NOETHER_INTERSLAVIC_EDITABLE_SOURCES.zip`;
3. `02_NOETHER_INTERSLAVIC_EVIDENCE_AND_PROVENANCE.zip`;
4. `03_NOETHER_INTERSLAVIC_SHA256_MANIFEST.txt`.

Prvy fajl jest ljudsky čitateljnik. Iztočny arhiv sodrži latinni kanon, izvedene kirilične izvorniky, projektor, testy, graditelje, metadane i členske manifesty. Dokazny arhiv sodrži redakcijny žurnal, do ktorogo se može samo dodavati, bočne zapisy rěšenj, polny manifest rolij, prověrčne dokumenty, tuta metodologična

statja s LaTeX-iztočnikom, QA i zadržane dokazy neuspěhov. Operacijske predpomněti, tajne ključe, surove razgovorne zapisy i nevezane rabotne oblasti ne sut javny dokazny produkt.

Stabilny koncept izdanja jest 10.5281/zenodo.21926382; verzijny DOI v020 jest 10.5281/zenodo.22058530; javny repozitorij jest KokunoYumeto/emmy-noether-isv. Opis v repozitoriju i metadane Zenodo stavjajut najprvo polny medžuslovjansky uvod, potom polnu anglijsku paralelnu verziju. Publikacija završaje se anonimnym povratnym čitanjem vsakego javnogo fajla i sravnjenjem bajtov i SHA-256 s lokalnym zamrznutym manifestom.

11 Ograničenja, ljudsky pregled i odgovorne tvrdženja

Izdanje može s uvěrjenostju tvrdžiti točny vhodny vektor, povtorimu redakcijnu historiju, polny pregled projektorskih rolij, deterministični kiriličny izvod, ograničenu izhodnu razliku, sgrađanja bez pogrěšek i prověrjenu PDF-strukturu. Ono ne tvrdži:

- rodno-govoriteljsku certifikaciju ili soglasje medžu slovjanskymi općinami;
- potvrđenje medžuslovjanskoj općiny ili oficialnogo standarda;
- nezavisnu naučnu recenziju;
- novu kritičnu redakciju německyh originalov;
- že računalna prověrka dokazuje idiomatičnost ili optimalny leksičny izbor;
- že kirilica jest nezavisny dokaz pravilnosti prěvoda.

Člověčsky vlastnik projekta je oprědělil obseg, kustodiju, dostup i publikacijne cěljy. Dostupny dokaz ne opravdava tvrdženje, že člověk ili rodny govoritelj je liniju-po-liniji recenzoval cěly jezyčny korpus. Slova “polny pregled” v v020 zato označajut završenu pojavjenjsko-rolnu klasifikaciju, izvedenu mašinno-pomaganym trudom i prověrjenu determinističnymi kontrolami. Ona ne značet ljudsku jezyčnu certifikaciju.

Ta ograničenja ne sut razlog skryti dělo. Javny izvorniky, lokatory i kontrolne sumy prětvarjajut nesoglasje v izvršivu popravku: recenzent može imenovati město, prědložiti lučše čitanje, citovati protivny dokaz i napraviti naslědno izdanje, ktore dodava popravku bez izbrisanja prědhodnogo stanja.

12 Povtorimy protokol za druge jezyky i transkripcije

Najmenši prěnosimy protokol jest:

1. oprěděliti jedin iztočny avtoritet i jedin ciljevy jezyčny kanon;
2. pridati eksplicitnu rolu vsakoj izvedenoj azbukě, transkripciji ili formatu;
3. zamrznuti iztočne jedinice s bajtami i hašami prěd redakcijeju;
4. zapisovati materialne rěšenja v žurnal, do ktorogo se može samo dodavati, zajedno s odklonjenymi alternativami;
5. razděliti vidimy jezyk od zaštićenoj sintakse, matematike i identitetov;
6. pregledati kandidaty po pojavjenju i vezati klasifikacije k stabilnym lokatoram;
7. prověriti smysl, jezyk, matematiku, sintaksu, sgrađanje, izteg, fonty i renderovanje kako različne vorota;
8. publikovati ljudsky čitateljnik, redaktabilne izvorniky, dokazy i vrhnji haš-manifest;
9. izložiti mašinno čitljiv indeks kanona i determinističke vhody;
10. javno priznati odsutny pregled i ostalu neuvěrjenost;
11. izdavati neměnljive verzije i anonimno prověriti javne bajty.

Glavna lekcija jest prosta: AI-stvorjeno jezyčno izdanje stavaje naučno uporabljivo togda, kogda njegovi izbori ne ostavajut v razgovoru modela. Kanon, role, dokazy, neuspěhi, komandy sgrađanja i javne identitety morajut živěti nezavisno od sesije, ktora jih je stvorila.

Citovanje, dostupnost i ponovno uporabjenje

Prědlagano citovanje: *Machine-assisted Interslavic editorial project. Od jezyčnogo kanona do povtorimoj publikacije: metodologija mašinno-pomaganoj medžuslovjanskoj redakcije / From a language canon to a reproducible publication: machine-assisted Interslavic editing of Emmy Noether's mathematical works. NOETHER-ISV-v020.* 10.5281/zenodo.22058530 (2026).

Tuta metodologična statja i druge projektom stvorjene redakcijne materialy sut ponudžene pod CC0 v měrě, v ktorej prava obstajajut. Iztočne děla i materialy tretjih stran zadrživajut svoj vlastny pravny status. LaTeX-iztočnik jest vključen, da tekst može byti popravljany, razširjany ili ponovno publikovany s točnym avtorstvom, verzijnoju identitetoju i odkrytym rodoslovjem dokazov.

Part II - Complete English counterpart

13 Object, purpose, and readership

The object is one complete Interslavic language edition of a bounded Emmy Noether corpus: numbered Papers 1–43, lecture work 44, Paper 45, and the bibliography. It is not an anthology of samples and not a collection of unrelated transcriptions. The public reader provides two writing surfaces of one language object:

- `isv-Latn` is the canonical editorial language text;
- `isv-Cyrl` is a deterministic projection of that text, not an independent translation witness.

This distinction is methodologically important. A derived script cannot itself confirm the translation from which it was created. Linguistic corrections enter only the Latin canon; Cyrillic must be regenerated, tested, built, and visually reviewed after the input bytes are frozen. V020 is an especially clear case: the Latin text remains byte-identical to v019, but the derived Cyrillic is corrected because the role review found wrong boundaries among language text, names, quotations, and Roman labels.

The practical heuristic for linguistic progress is not "how Slavic-looking is the prose?" It is the conjunction of four questions:

1. Can speakers of different Slavic languages plausibly parse the sentence without English?
2. Is the selected form supported by the declared Interslavic lexicon or grammar, or by a clearly recorded project-house rule?
3. Does the form preserve the exact mathematical referent, quantification, and argument relation of the German source?
4. Is every exception, identity, quotation, or ambiguity represented by a reproducible role rather than silently omitted?

The edition is therefore best understood as a corrigible scholarly working edition: public, inspectable, hash-bound, and open to improvement, but not represented as native-speaker-certified or as a critical edition.

14 Authority and custody architecture

The method separates source authority, language canon, derived script, editorial evidence, and public products. No layer may silently assume the function of another.

The German object inherited from v019 has SHA-256 `51A25101C04877AE740989E72B2AD65A7A7E65B081077C4A518BF1737AD5B907`. This identity marks the control source against which the Latin canon was frozen; it is not a claim that this project published a new critical German text. The global multilingual collection is related by `IsPartOf` to 10.5281/zenodo.20412587, and the identified German object by the semantic relation `IsTranslationOf`.

15 Machine-readable canon discovery

Repeatability begins with discovery. A future session must not infer which of several similar files is canonical. The repository root therefore exposes `CANON_INDEX.json`, which declares:

- language and BCP 47 script tags;
- the four canonical Latin files and four derived Cyrillic files;
- exact byte counts and SHA-256 values for every controlling unit;
- the Latin editorial head and append-only ledger identity;
- the role manifest, projector, test, and validation receipt;
- German authority, write boundaries, and claim boundaries;
- deterministic build commands and public artifact names.

Table 4: Authority roles in NOETHER-ISV-v020.

Role	Controlling object	Consequence
German source authority	Inherited object `NOETH-DE-ED-0015`, pointer `NOETH-DE-AUTH-v052-20260815`	Controls source sense, structure, and mathematical comparison; the Interslavic lane does not edit it.
Latin language canon	Four `source_latin/*.tex` files at head `ISV019-EDIT-0171`	The only surface for linguistic edits. V020 changes zero of its bytes.
Role manifest	`ISV020-REVIEWED-PROJECTION-ROLE-MANIFEST.json`	Binds 4,560 reviewed records, 8 additional boundaries, and zero unresolved classifications to exact source locators.
Cyrillic surface	Projector v7 over the exact Latin vector and manifest	Reproducibly generated output; cannot overrule the Latin canon or independently validate the translation.
Release products	Two readers, linked reader, two ZIP archives, and SHA-256 manifest	Human-facing dissemination; products cannot redefine the canon that produced them.

The index is declarative, not self-authenticating. It tells an agent what the objects *are*; executable checks prove whether the declared identities still hold. This separation rejects the circular rule "the newest-looking file is the canon" and provides a base to which another transcription, translation, or script projection can safely add a new layer.

16 What AI does and what it cannot decide alone

The project uses AI as an editorial instrument, not as source authority. Machine work includes sentence-level comparison, candidate inventory construction, occurrence classification, formulation of alternatives, generation of structural tests, and production of public documentation. Deterministic programs then authenticate inputs, apply only frozen decisions, and reject unintended changes.

The method follows this cycle:

1. freeze source and target byte vectors;
2. locate material linguistic or role candidates without automatic global replacement;
3. review every occurrence in sentence, bibliography, mathematics, and TeX context;
4. record the chosen role, evidence, rejected alternatives, uncertainty, and consequences;
5. produce a manifest or editorial decision bound to an exact locator and hash;
6. derive output with a pure program and prove the allowed delta;
7. build, extract text, inspect fonts, navigation, structure, and rendered pages;
8. publish sources, evidence, and hashes with the reader, then anonymously read the public bytes back.

An AI proposal does not become canonical because it sounds plausible. It becomes admissible only when localized, supported by declared evidence, bounded in scope, and reproducibly verified. Where evidence is insufficient, the proper output is a narrow claim or a recorded open question, not invented certification.

17 Complete role review and the v020 correction

V019 already had a functional Cyrillic projection, but its inherited inventory originated in candidate scanners that can mistake a foreign-looking token, styled text, or a single letter for material that should automatically be preserved. V020 therefore reviewed all 4,560 records occurrence by occurrence, not only obvious defects.

In addition to the 364 corrected classifications, the review added 8 missing spans for complete uninflected

Table 5: Role-classification review.

Category	Records reviewed	Classifications corrected
Styled surfaces	883	1
Name candidates	616	0
Foreign-lexicon occurrences	2,371	270
Visible one-letter tokens	690	93
Total	4,560	364

identities. There are therefore 372 bound effects. Projector v7 checks each effect independently: 346 ranges must preserve Latin material and 26 must project it. Zero effects remain unverified.

The change licence is not a free list of whole files. The manifest computes 309 lines on which a reviewed effect could lawfully change output. The actual delta contains 125 lines: 0 in work 44, 6 in paper 45, 115 in papers 1–43, and 4 in the bibliography. Zero lines change outside the licensed set. The four Latin sources have zero changed bytes.

The status “zero ambiguity” must be read operationally: every inventory record has one completed projection role and an exact locator. It does not assert that natural language has no polysemy, that every speaker would choose the same form, or that a native speaker reviewed the complete corpus.

18 Append-only linguistic normalization of the Latin canon

The Latin head inherits 171 material decisions from v019. A decision is admissible only when it identifies a predecessor, exact target locus, proposed successor, relevant authority, rejected alternatives, motivation, uncertainty, and post-application validation. An applicator authenticates every input and rejects ambiguous targets. The ledger is append-only: a failed or superseded approach remains readable with its successor instead of disappearing.

The main linguistic classes are:

- **False friends and semantic partitions.** Resemblance to Russian, Ukrainian, Polish, Czech, or South Slavic forms is insufficient. Existence, essence, composition, containment, assumption, consequence, and correspondence are adjudicated by sentence and mathematical sense.
- **Morphology and sentence role.** Agreement, oblique cases, genitive plurals, participles, verb endings, and ordinal forms are classified before correction. Global suffix replacement is rejected because formulas, names, and quotations can imitate an ordinary morpheme.
- **Connectives and proof logic.** Causal, inferential, conditional, equivalence, contrastive, and apodosis functions are compared with the relation between source clauses. Word frequency must not reverse implication or weaken a quantifier.
- **Names, titles, and quotations.** A complete uninflected identity and an original-language bibliographic surface are preserved. An integrated or inflected form is projected when it functions as Interslavic morphology.

V020 does not reopen those Latin decisions. It corrects only the machine interpretation of which portions of the frozen text are visible Interslavic and which are protected identity or structure.

19 Mathematical and TeX fidelity

Natural-language normalization is subordinate to mathematical identity. Protected boundaries include math mode, command names, command options, labels, references, citation keys, paths, comments, and demonstrated original-language arguments. For every Latin/Cyrillic pair, removing Unicode letters and combining marks yields the same non-letter stream. This check preserves braces, control words, operators, numbers, alignment marks, and mathematical symbols.

An earlier failure demonstrates why a linguistic scanner is insufficient. Projector v5 misread the TeX unit `24em` as a visible word and created a Cyrillic unit that is not legal TeX. V6 therefore acquired an explicit grammar for numeric dimensions, which v7 inherits and rechecks. It preserves 447 previously exposed occurrences: 219 `em` and 228 `pt`; Cyrillic dimension-unit residue is zero. The failure remains in the evidence because a clean lexical result does not prove that the source program is executable.

20 Deterministic v7 derivation

V7 is a pure function of the four frozen Latin files, the reviewed role manifest, and pinned dependencies. It:

1. authenticates every Latin unit by byte count and SHA-256;
2. verifies exact source and manifest topology;
3. parses TeX into visible and protected roles;
4. merges reviewed boundaries for names, quotations, Roman labels, and dimensions;
5. projects every remaining visible word with the frozen Interslavic transliterator;
6. independently reconstructs the output with a trace and requires byte identity between both derivations;
7. proves every one of the 372 effects and every changed line;
8. writes output transactionally only after every gate passes.

Hostile tests deliberately mutate a source byte, omit a topology member, propose an unsafe output target, corrupt the manifest pin, and corrupt an old-output pin. All five cases are rejected before writing. Two independent calculations return byte-identical sets of four Cyrillic outputs. The test and projector are themselves hashed, so “the same procedure” means identical code, not merely a similar description.

21 Build and multilayer validation

All XeLaTeX operations run serially, with two passes per document, shell escape disabled, and a fixed `SOURCE_DATE_EPOCH`. Eight editable components produce two complete readers.

Table 6: Verified v020 component readers.

Artifact	Pages	Bytes	SHA-256
Latin reader	565	3,000,522	`B27278EA138724ED258CFE5089CABA70A79C640D0B71CCD9836CA3C8CB8BCD13`
Cyrillic reader	588	3,300,221	`4AE9847672AB53F1D5B54468ED9B202D45EF5969C7A34B1CE380B28C6516DB29`

Release QA reports 38 passing checks and zero errors. It covers exact source readback, TeX and non-letter structure, fatal logs, page topology, fonts, `pdftotext`, script census, and text identity between component and cumulative readers. The linked public reader must contain 6 front/divider pages plus 565 Latin and 588 Cyrillic pages, for a total of 1,159. Its builder rejects any other topology.

A structural test does not prove good appearance. The final release gate therefore also requires serial rendering and visual review of the new front matter, every changed page and adjacent control pages, plus the boundaries between scripts. Text extraction alone cannot reveal overlap, clipping, or an unreadable glyph substitution.

22 Publication, access, and preservation

The public Zenodo surface contains four products:

1. `00_NOETHER_INTERSLAVIC_COMPLETE_LINKED_READER.pdf`;
2. `01_NOETHER_INTERSLAVIC_EDITABLE_SOURCES.zip`;

3. `02_NOETHER_INTERSLAVIC_EVIDENCE_AND_PROVENANCE.zip`;
4. `03_NOETHER_INTERSLAVIC_SHA256_MANIFEST.txt`.

The first file is the human reader. The source archive contains the Latin canon, derived Cyrillic sources, projector, tests, builders, metadata, and member manifests. The evidence archive contains the append-only editorial ledger, decision sidecars, complete role manifest, validation receipts, this methods paper and its LaTeX source, QA, retained adverse evidence, and its own member manifest. Operational caches, credentials, raw conversation records, and unrelated workspaces are not public evidence products.

The stable edition concept is 10.5281/zenodo.21926382; the v020 version DOI is 10.5281/zenodo.22058530; the public repository is KokunoYumeto/emmy-noether-isv. The repository introduction and Zenodo metadata place a complete Interslavic presentation before a complete English counterpart. Publication concludes with anonymous readback of every public file and comparison of its bytes and SHA-256 with the locally frozen manifest.

23 Limitations, human review, and responsible claims

The edition can make strong claims about an exact input vector, replayable editorial history, complete projector-role review, deterministic Cyrillic derivation, bounded output delta, clean builds, and verified PDF structure. It does not claim:

- native-speaker certification or consensus among Slavic communities;
- endorsement by the Interslavic community or an official standard;
- independent scholarly peer review;
- a new critical edition of the German originals;
- that computational validation proves idiomaticity or an optimal lexical choice;
- that Cyrillic is independent evidence for the translation.

A human project owner set scope, custody, access, and publication goals. The available evidence does not justify claiming that a human or native speaker reviewed the entire language corpus line by line. “Complete review” in v020 therefore denotes completed occurrence-level role classification performed with machine assistance and verified by deterministic controls. It does not denote human linguistic certification.

These limitations are not reasons to hide the work. Public sources, locators, and hashes turn disagreement into an executable correction: a reviewer can identify a locus, propose a better reading, cite contrary evidence, and produce a monotonic successor without erasing the prior state.

24 Reusable protocol for other languages and transcriptions

The minimum transferable protocol is:

1. define one source authority and one target-language canon;
2. assign an explicit role to every derived script, transcription, or format;
3. freeze source units with byte counts and hashes before editing;
4. record material decisions append-only with rejected alternatives;
5. separate visible language from protected syntax, mathematics, and identities;
6. review candidates occurrence by occurrence and bind classifications to stable locators;
7. validate meaning, language, mathematics, syntax, builds, extraction, fonts, and rendering as distinct gates;
8. publish a human reader, editable sources, evidence, and a top-level hash manifest;
9. expose a machine-readable canon index and deterministic entry points;
10. disclose absent review and remaining uncertainty;
11. issue immutable versions and anonymously verify public bytes.

The central lesson is simple: an AI-produced language edition becomes scholarly usable when its choices are not trapped in a model conversation. The canon, roles, evidence, failures, build commands, and public identities must survive independently of the session that created them.

Citation, availability, and reuse

Suggested citation: *Machine-assisted Interslavic editorial project. Od jezyčnogo kanona do povtorimoj publikacije: metodologija mašinno-pomaganoj medžuslovjanskoj redakcije / From a language canon to a reproducible publication: machine-assisted Interslavic editing of Emmy Noether's mathematical works. NOETHER-ISV-v020.* 10.5281/zenodo.22058530 (2026).

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